End-Term Examination
(CBCS) (SUBJECTIVE TYPE)
B. Tech CSE AI 4th Semester
(May, 2022)

Q.P Code 220501

Subject Code: BAI 202	Subject: Computer Networks
Time : 3 Hours	Maximum Marks : 60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1.		**(2.5*8=20)**
	(a) What is Shannon capacity? Explain	
	(b) How are computer networks classified based on geography?	
	(c) What are LDPC codes?	
	(d) What is subnetting?Write the subnet mask of class C in bits	
	(e) What is NAT?	
	(f) Explain OSPF protocol briefly.	
	(g) What is CIDR notation? Explain	
	(h) Give the port number for UDP	
	UNIT-1	
Q2.	Identify the layers in OSI reference model and illustrate their functions.	**(10)**
Q3.	Describe the role of network layer and transport layer in TCP/IP model	**(10)**
	UNIT-2	
Q4.	Explain CRC, LRC and VRC along with relevant examples.	**(10)**
Q5.	Describe CSMA/CD protocol in Ethernet.	**(10)**
	UNIT-3	
Q6.	Compare and contrast Link State routing and Distance vector routing	**(10)**
Q7.	Compare and contrast IPV4 header with IPv6 header	**(10)**
	UNIT-4	
Q8.	Explain three way handshaking protocol with relevant diagrams	**(10)**
Q9.	Explain the leaky bucket algorithm for traffic shaping	**(10)**